# Test 2 review day

## CMSE 381

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Fri, March 14, 2025

# Announcements

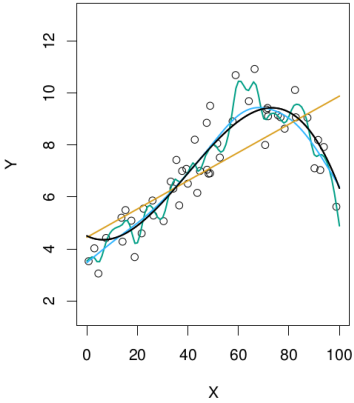
Announcements:

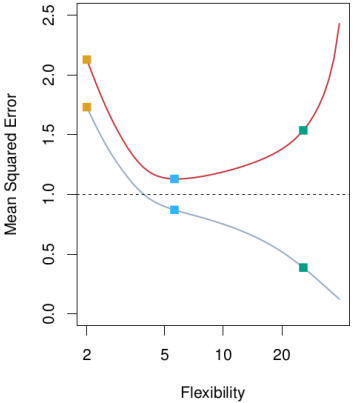
- midterm # 2 on Monday
- HW 5 due Sunday (3/16)

# Top questions submitted

- Compare different kinds of validation
- difference between lasso and ridge regression
- go through subset selection example again

# Training error vs test error

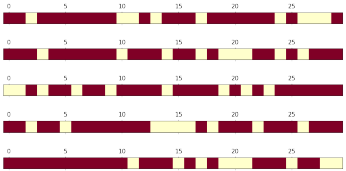


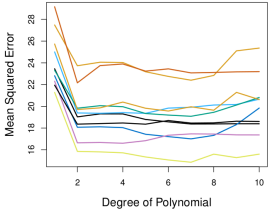




- Left: Black line is model for simulation of data
- Right side: training in blue/grey, testing in red
- Variance of $\varepsilon$ is dashed line
- Point out that training error goes down but test goes up (overfitting)
- Flexibility = degrees of freedom

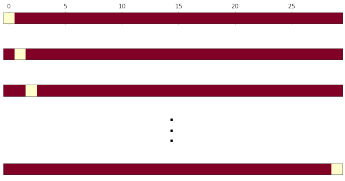
# Approximations of Test Error

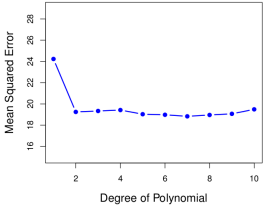
## Validation Set



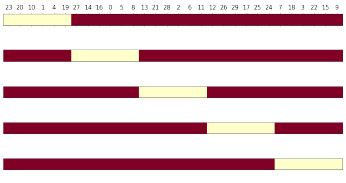


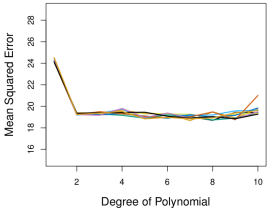

## LOOCV



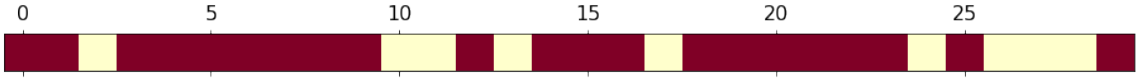




## K-fold CV



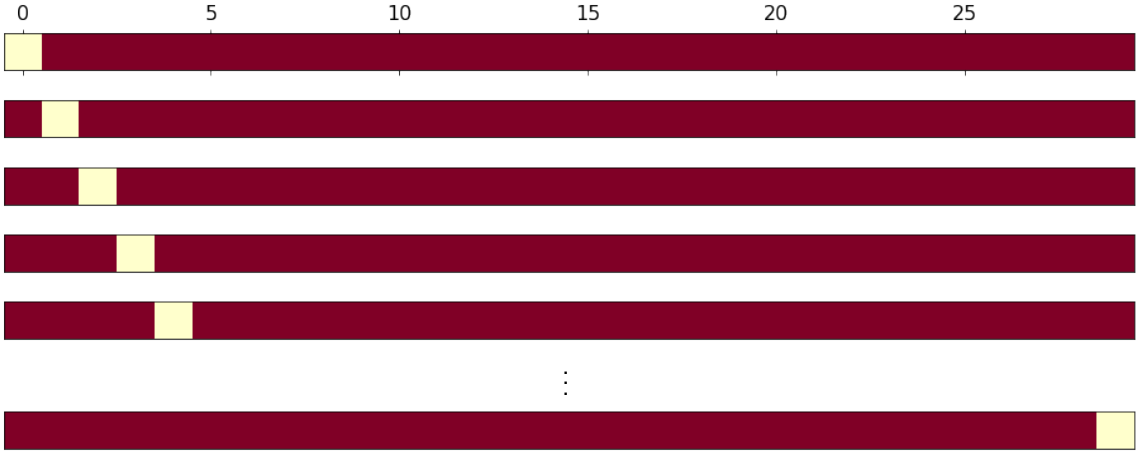
# Validation set approach




- Divide randomly into two parts:
  - Training set
  - Validation/Hold-out/Testing set
- Fit model on training set
- Use fitted model to predict response for observations in the test set
- Evaluate quality (e.g. MSE)

- This example has 30 data points
- Keep out 30% = 9 for testing in yellow
- We've done this in previous jupyter notebooks

# LOOCV



# LOOCV in mathy words

- Remove $(x_1, y_1)$ for testing.
- Train the model on $n-1$ points: $\{(x_2, y_2), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_1 = (y_1 - \hat{y}_1)^2$

- Remove $(x_2, y_2)$ for testing.
- Train the model on $n-1$ points: $\{(x_1, y_1), (x_3, y_3), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_2 = (y_2 - \hat{y}_2)^2$

- Rinse and repeat

Return the score:

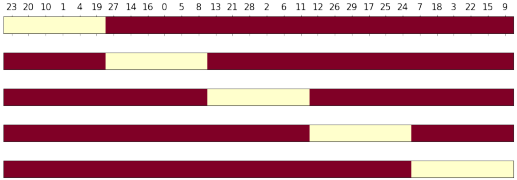
$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i$$

# Definition of $k$-fold CV

- Randomly split data into $k$-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has $\ell$ points

- Remove $i$th fold $U_i$ and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell}\sum_{(x_j,y_j)\in U_i}(y_j - \hat{y}_j)^2$

- Rinse and repeat




Return

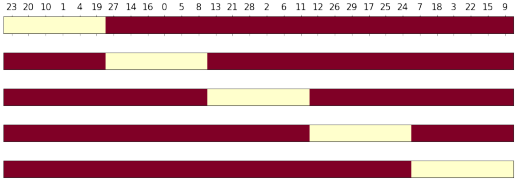
$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k}\mathrm{MSE}_i$$

# Setup: $k$-fold

- Randomly split data into $k$-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has $\ell$ points

- Remove $i$th fold $U_i$ and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{Err}_i = \frac{1}{\ell} \sum_{(x_j, y_j) \in U_i} \mathrm{I}(y_j \neq \hat{y}_j)$

- Rinse and repeat




Return

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^{k} \mathrm{Err}_i$$

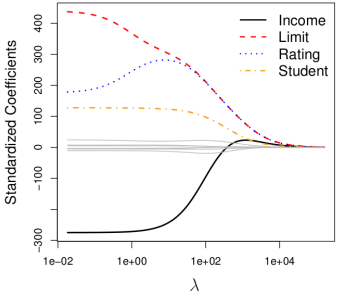
# Ridge regression

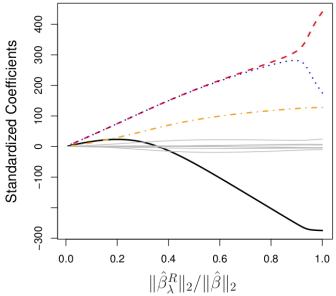
**Before:**

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

**After:**

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$



# The lasso

**Least Squares:**

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

**Ridge:**

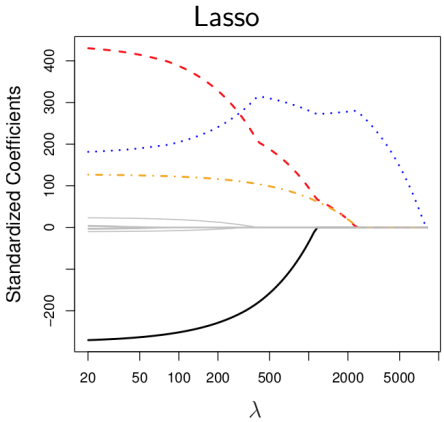
$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$

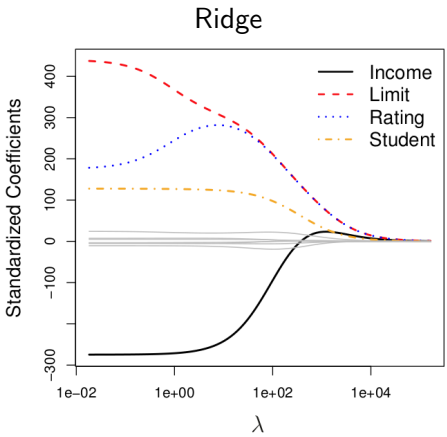
**The Lasso:**

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\sum_{j=1}^{p}|\beta_j|$$

- $\ell_1$ penalty instead of $\ell_2$ penalty
- $\ell_1$ norm: $\|\beta\|_1 = \sum|\beta_j|$
- $\ell_2$ norm: $\|\beta\|_2 = \sum\beta_j^2$
- $\ell_p$ norm: $\|\beta\|_p = (\sum|\beta_j|^p)^{1/p}$
- $\ell_\infty$ norm: $\|\beta\|_\infty = \max\{|\beta_i|\}$

# An example on `Credit` data set



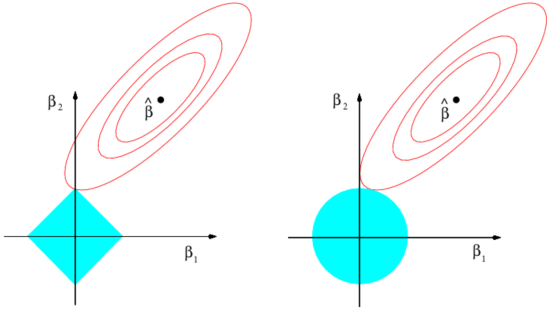

- $\lambda = 0$: Both are least squares
- $\lambda = \infty$: Everything is 0
- only lasso has 0 efficients at finite $\lambda$

# Why the hell lasso can select variable (while ridge cannot)?

Alternative formulation of lasso & ridge regression (play more with $\ell_p$)

$$\min_{\beta} \sum (y_i - \hat{y}_i)^2 \text{ where } \sum |\beta_j| \leq s$$

$$\min_{\beta} \sum (y_i - \hat{y}_i)^2 \text{ where } \sum |\beta_j|^2 \leq s$$



# Subset selection example: best subset

We train a model using four variables, $X_1, X_2, X_3$. We're interested in getting a subset of the variables to use. The following table shows the training and test error for the model learned using each possible subset of variables.

| | Training MSE | K-fold CV test MSE |
|---|---|---|
| null model | 0.079 | 0.614 |
| X1 | 0.984 | 0.954 |
| X2 | 0.172 | 0.642 |
| X3 | 0.410 | 0.533 |
| X1,X2 | 0.800 | 0.810 |
| X2,X3 | 0.696 | 0.577 |
| X1,X3 | 0.888 | 0.779 |
| X1,X2,X3 | 0.133 | 0.536 |

# Subset selection example: forward selection

We train a model using four variables, $X_1, X_2, X_3$. We're interested in getting a subset of the variables to use. The following table shows the training and test error for the model learned using each possible subset of variables.

| | Training MSE | K-fold CV test MSE |
|---|---|---|
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| X1,X2 | 0.800 | 0.810 |
| X2,X3 | 0.696 | 0.577 |
| X1,X3 | 0.888 | 0.779 |
| X1,X2,X3 | 0.133 | 0.536 |

# Subset selection example: backward selection

We train a model using four variables, $X_1, X_2, X_3$. We're interested in getting a subset of the variables to use. The following table shows the training and test error for the model learned using each possible subset of variables.

| | Training MSE | K-fold CV test MSE |
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